# P09

## Network Tomography: Estimating Source-Destination Traffic Intensities from Link Data


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| Domain | Signal processing, statistics & applied mathematics |


### Abstract (quoted verbatim from the publication)

The problem of estimating the node-to-node traffic intensity from repeated measurements of traffic on the links of a network is formulated and discussed under Poisson assumptions and two types of traffic-routing regimens: deterministic (a fixed known path between each directed pair of nodes) and Markovian (a random path between each directed pair of nodes, determined according to a known Markov chain fixed for that pair). Maximum likelihood estimation and related approximations are discussed, and computational difficulties are pointed out. A detailed methodology is presented for estimates based on the method of moments. The estimates are derived algorithmically, taking advantage of the fact that the first and second moment equations give rise to a linear inverse problem with positivity restrictions that can be approached by an EM algorithm, resulting in a particularly simple solution to a hard problem. A small simulation study is carried out.

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2. What is unknown or hidden, and what can actually be observed or measured?
3. What method do the authors use? Is it new, borrowed from another paper, or a modification of one?
4. What evidence do they give (proofs, simulations, experiments, real data)?
5. What assumptions or limitations can you find?
6. Which other papers in the corpus does it cite, use, extend or resemble? Note the page or section.

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